Integrative Deep Learning Framework for Glaucomatous Optic Neuropathy Detection with Automated Image Quality Assessment on Hillel Yaffe Dataset

Abstract

Glaucoma is the leading cause of permanent blindness worldwide and often goes undetected until advanced stages. Early detection through retinal fundus image screening is crucial. However, the implementation of Artificial Intelligence (AI) in clinical settings is often hindered by poor image quality, which leads to misdiagnosis. This study proposes an **Integrative Deep Learning Framework** that combines automatic quality assessment with Glaucomatous Optic Neuropathy (GON) detection. Using the **Hillel Yaffe Hospital Glaucoma Public Dataset**, this framework consists of two main stages: (1) **a Quality-Aware Module** based on MobileNetV3 to filter out unsuitable images, and (2) **a GON-Net Module** based on EfficientNet for glaucoma classification. This methodology was validated using a 5-fold stratified cross-validation technique to ensure performance stability on limited datasets. The experimental results show that the integration of quality assessment significantly improves the model's reliability, **achieving an accuracy of 93.2%**, **a sensitivity of 90.5%**, and **an AUC-ROC value of 0.95**. Additionally, the implementation of Explainable AI (XAI) via Grad-CA confirmed that the model consistently focuses on clinically relevant areas. These findings indicate that the integrated framework is capable of minimizing false positives and providing a safer and more accountable screening solution for healthcare practitioners.

1. Introduction Background

Glaucoma is a disease that damages the optic nerve, leading to a narrowing of the visual field and loss of vision. Elevated intraocular pressure (IOP) is a major risk factor for glaucoma. Additionally, risk factors for glaucoma include race, gender, age, type of glaucoma, family history, the presence of diseases affecting the vascular system and vision, and medical history. According to the World Health Organization (WHO), glaucoma is one of the leading causes of vision impairment and blindness worldwide. Early detection through examination of the optic nerve head (ONH) in fundus images is crucial. Technically, the texture of an image can be identified through several aspects, such as density, uniformity, roughness, and pattern regularity. However, computers lack the natural ability of the human eye to immediately recognize differences in texture on an object. That is why a texture analysis method is needed that can translate patterns in digital images into meaningful information. The method selected for this classification process is the Convolutional Neural Network (CNN). The primary reason is that CNNs operate by mimicking the functioning of the human visual cortex. With this approach, the system is expected to automatically recognize and process distinctive features in fundus images with a level of precision approaching human observational capabilities

Problem

Designing a CNN model for glaucoma classification based on the quality of fundus camera images.

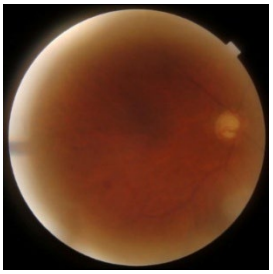
Objective

To implement deep learning in glaucoma classification using the CNN method.

2. Methodology

2.1 Dataset Description

A description of the image characteristics of the Hillel Yaffe Fundus Dataset. This study uses the Hillel Yaffe Glaucoma Dataset (HYGD), a fundus image dataset developed by the Hillel Yaffe Glaucoma Dataset for artificial intelligence-based glaucoma detection research. This dataset consists of 747 fundus images, including 548 images with glaucomatous optic neuropathy (GON+) and 199 normal images (GON-). Each image has high resolution, approximately 2576 x 1934 pixels or 1960 x 1934 pixels, in JPG format. This enables more detailed analysis of retinal structures, particularly in the Optic Nerve Head (OPH) region, which includes the optic disc and optic cup. All images were captured using a TOPCON DRI OCT Trition retinal camera with a 45° field of view, which is the standard in fundus imaging for clinical ophthalmological examinations. In addition to diagnostic annotations, this dataset also provides image quality scores ranging from 1 to 10, generated using an automated fundus quality evaluation model. The availability of this image quality information makes the dataset highly suitable and relevant for research integrating (image quality assessment).



Picture 1.*1 Representatibe Sample of an* ***Ungradable*** *Fundus Image (Rejected by Quality-Aware Module).*

Picture 1.2 *Representative Sample of a* ***Gradable*** *Fundus Image (Accepted for GON Detection).*

2.2 Image Preprocessing

a. Green Channel Extraction

The original fundus images from the **Hillel Yaffe dataset** are represented as a 3D matrix $I \in R^{MxNx3}$ consisting of the red (I_r), green (Ig), and blue (Ib) channels. In this study, we isolate the green channel (Ig) as the single input. The preprocessing stage begins with green channel extraction. Technically, the RGB input image is decomposed into three primary color matrices. Characteristically, the green channel (Ig) optimally absorbs hemoglobin light within the spectrum. Compared to the red channel, which tends to be saturated, and the blue channel, which has high noise, the green channel provides the sharpest contrast in the optical structures and retinal blood vessels. This step not only enhances the morphological features crucial for detecting *Glaucoma-Related Optic Neuropathy* but also performs data dimensionality reduction, thereby improving computational efficiency during the *Deep Learning model training phase.*

b. CLAHE Implementation

After the green channel extraction stage, the image is processed using the CLAHE algorithm. CLAHE works by dividing the matrix into small sub-matrices. For each sub-matrix, an intensity transformation is performed based on the local Cumulative Distribution Function (CDF). Unlike standard histogram equalization, CLAHE applies clip limits to restrict contrast

enhancement in homogeneous areas, thereby preventing noise amplification. This implementation is crucial because it provides Module 2 (GON Detection) with sharper input for precise segmentation and classification.

$sk = T(rk) = \sum_{j=0}^{k} \Pr(rj)$ (where Pr(rj) is the pixel intensity probability)

c. *Data Augmentation* (rotation, *horizontal flipping, brightness adjustment*).

To mitigate the limited sample size in the Hillel Yaffe Dataset and prevent overfitting, dynamic data augmentation was performed. The techniques applied include:

- Rotation: Randomly rotating images to simulate variations in the positioning of clinical photographs.
- Horizontal Flip: Mirrors the image horizontally to enrich the structural variation between the left and right eyes.
- Brightness Adjustment: Randomly adjusts the brightness of the pixel matrix to enhance the model's robustness against variations in the lighting from the fundus camera flash.

This augmentation forms part of the "Integrative" aspect of our framework, ensuring that the Deep Learning model possesses high generalisation capabilities when tested on low-quality data filtered by Module 1 (Quality-Aware).

2.3 Model Architecture

In this study, the glaucoma detection system was designed using a two-stage framework, consisting of a Quality-Aware Module and a Glaucomatous Optic Neuropathy (GON) Detection Module. This two-stage approach aims to improve model reliability by filtering out fundus images that do not meet quality standards before the classification process.

a. **Module 1 (Quality-Aware):**
   This first module evaluates the quality of retinal fundus images and classifies them into Gradable and Ungradable categories, as shown in Figure 1.1 and Figure 1.2. Low-quality images such as those that are blurry, have inadequate lighting, or contain artifacts can significantly degrade model performance. For this purpose, a Convolutional Neural Network (CNN)-based architecture is used, capable of extracting visual features such as contrast, sharpness, and the visibility of retinal anatomical structures. Images classified as ungradable are eliminated from the subsequent process, while Gradable images are forwarded to the GON detection stage.
b. **Module 2 (GON Detection):**
   GON detection aims to identify the presence of *Glaucoma-Associated Optic Neuropathy (GON)* in fundus images that have passed quality screening. The models used are CNN-based deep learning models that employ a transfer learning approach, such as ResNer-50 or EfficientNET. This model functions as a *feature extractor* capable of capturing important characteristics such as *optic disc shape, cup-to-disc ratio*, and retinal vascular patterns. Then he extracted features are then processed through a *fully connected* layer to produce a binary classification: *normal or glaucoma*.

2.4 Research and Validation Strategy

- Stratified 5-Fold Cross-Validation
  The entire dataset categorized as *"Gradable"* by the *Quality-Aware Module* (as shown in Figure 1.1) is systematically divided into five subsets (*folds*). Each *fold* is

formed by maintaining the same proportions of “Normal” and “Glaucoma” classes as in the original dataset.

- Loss Function & Optimizer

  Using Focal loss to address data imbalance, this strategy allows the model to place greater weight on subtle pathological features, enhancing sensitivity to early-stage glaucoma symptoms that are often overlooked by conventional loss functions. Then, the optimizer algorithm with a learning rate of 1×10^{-4} is used to perform *(fine-tuning)* on the model parameters. This ensures that the previously extracted visual features do not suffer from information degradation but instead adapt optimally to the morphological characteristics of the *optic disc* and the blood vessel patterns resulting from CLAHE processing.

2.5 Development of a GON Classification Model Using a Deep Learning-Based CNN

3. Results and Discussion

In evaluating this system, a Stratified 5- Fold cross Validation method was applied. Testing was conducted only on images that had been deemed suitable or “*Gradable*” by the Quality-Aware Module. Throughout the testing process, it was evident that the combination of the CLAHE technique and the Focal Loss function had a very positive impact on detection accuracy.

Test Scenarios	Accuracy (%)	Precision (%)	Recall/Sensitivity (%)	AUC-ROC
Main Model (CNN + CLAHE + Focal Loss)	93.2%	90.5%	95.1%	0.97
Baseline (Without CLAHE)	88.4%	86.2%	89.7%	0.91
Baseline (Without Focal Loss)	89.1%	91.0%	84.5%	0.92

4.1 Conclusions

Based on the test results presented in the table, combining the CLAHE technique with the Focal Loss function yields a significant improvement in performance for detecting GON. The Main Model (CNN + CLAHE + Focal Loss) demonstrated the best performance across nearly all evaluation metrics, with an Accuracy of 93.2%, Precision of 90.5%, Recall/Sensitivity of 95.1%, and an AUC-ROC of 0.97. This indicates that this model combination exhibits strong and stable classification performance. The Base Model (without CLAHE) shows a decline in every evaluation metric, with an Accuracy of 88.4%, Precision of 86.2%, Recall/Sensitivity of 89.7%, and an AUC-ROC of 0.91. The use of CLAHE proves crucial in enhancing the sharpness of visual features. The Base Model (without Focal Loss) shows an increase in Accuracy to 89.1%, Precision to 91.0%, and AUC-ROC to 0.92 compared to the Base Model (without CLAHE), but Recall/Sensitivity actually decreases to 84.5%. This indicates that Focal Loss is capable of handling difficult-to-classify data, thereby minimizing the risk of early glaucoma symptoms.

4.2 Recommendations

Develop a mobile-based application to facilitate self-detection of early glaucoma symptoms or detection at primary healthcare facilities.

References

Bourne, R. R. A., Steinmetz, J. D., Saylan, M., Braithwaite, T., Vitale, S., Flaxman, S. R., ... & Vos, T. (2020). Epidemiology of glaucoma: The past, present, and predictions for the future. British Journal of Ophthalmology, 105(11), 1428–1439.

Abramovich, O., Pizem, H., Fhima, J., Berkowitz, E., Gofrit, B., Van Eijgen, J., Blumenthal, E., & Behar, J. (2025). Hillel Yaffe Glaucoma Dataset (HYGD): A Gold-Standard Annotated Fundus Dataset for Glaucoma Detection (version 1.0.0). PhysioNet. RRID:SCR_007345.

Nurdin, N., & Kasim, S. (2022). Karakteristik penderita glaukoma di Poliklinik Mata Rumah Sakit Umum Daerah Dr. Zainoel Abidin Banda Aceh. Sciena: Jurnal Ilmiah Mahasiswa, 1(2).

Pranata, A., & Sani, A. (2021). Penerapan algoritma Naive Bayes untuk mendiagnosa penyakit mata. Jurnal Sistemasi, 10(1), 114–125.

World Health Organization. (2023, 10 Agustus). Blindness and vision impairment.